



H3Africa tranSMART Quick Start Guide

Box Plot with ANOVA



A Box Plot with ANOVA analysis depicts how a categorical variable (e.g. Disease) differs across a continuous variable (e.g. Age).

Step 1: Select samples

Drag and drop terms of interest into the Subset boxes on the **Comparison** tab. Include all terms to be used in the analysis.

Click on the **Summary Statistics** tab to review the number of samples that will be included in the analysis. Confirm that the set of samples available is sufficient for running a meaningful analysis.

Step 2: Set up analysis

Click on the **Advanced Workflow** tab and select “Box Plot with ANOVA” from the Analysis dropdown list.

Drag terms from the Navigate Concepts tree into the Independent Variable (1) and Dependent Variable (2) boxes. You can add multiple terms, or drag over a folder to add all the terms within it. One box should contain a continuous variable (e.g. Age) and the other box should contain a categorical variable (e.g. Disease).

Click the “Run” button (3). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses may take several minutes to run.

Active Filters: **AND** Clear

Comparison Summary Statistics Advanced workflow Workspace SmartR

Analysis: Box Plot with ANOVA

Save to

Cohorts: Subset 1: INCLUDE (\Private Studies\H3Africa_Cohort_v1\02 Participant Assignment\Confirmed\Participant Assignment: CKD or GN) AND INCLUDE (\Private Studies\H3Africa_Cohort_v1\01 Demographics\Age)

Variable Selection

Independent Variable
Select a variable from the Data Set Explorer Tree and drag it into the box. At least one of the variables selected should be a continuous variable (e.g. Age) and one should be a categorical variable (e.g. Tumor Stage). A continuous variable can be categorized using the binning option below.

Dependent Variable
Select a variable from the Data Set Explorer Tree and drag it into the box. At least one of the variables selected should be a continuous variable (e.g. Age) and one should be a categorical variable (e.g. Tumor Stage). A continuous variable can be categorized using the binning option below.

1 Chronic Kidney Disease (CKD)
2 Age (3327)

High Dimensional Data Clear High Dimensional Data Clear

Enable binning

3 Run

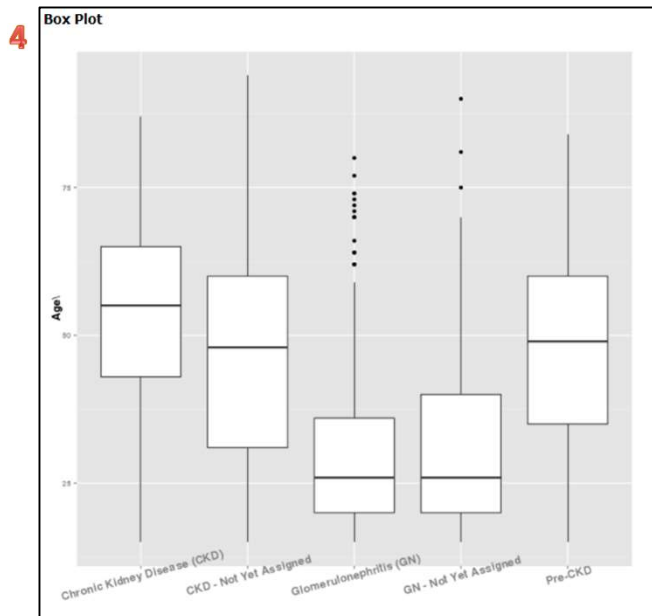
Navigate Concepts

- Private Studies
 - H3Africa_Cohort_v1 (3329)
 - 00 Study Disposition (3329)
 - 01 Demographics (3329)
 - Education (3309)
 - Employment and Income (3310)
 - Father of Participant Country of Birth (3311)
 - Marital Status (3310)
 - Mother of Participant Country of Birth (3311)
 - Participant Country of Birth (3311)
 - Sex (3316)
 - Age (3327)
 - 02 Participant Assignment (3283)
 - Confirmed (3061)
 - Infection Status (2571)
 - Participant Assignment: CKD or GN (3058)
 - Chronic Kidney Disease (CKD) (1654)
 - CKD - Not Yet Assigned (458)
 - Glomerulonephritis (GN) (516)
 - GN - Not Yet Assigned (206)
 - Pre-CKD (224)

Step 3: Review results

Results will appear underneath the variable selection boxes. The Independent variable will appear along the X-axis and the Dependent variable will be the Y-axis (4).

Scroll below the box plot to see ANOVA results (5).



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ANOVA Result

p-value	2.33e-207
F value	283

Group	Mean	n
Chronic Kidney Disease (CKD)	53.4	1654
CKD - Not Yet Assigned	46.1	457
Glomerulonephritis (GN)	30.0	515
GN - Not Yet Assigned	31.8	206
Pre-CKD	47.3	224

Pairwise t-Test p-Values

	Chronic Kidney Disease (CKD)	CKD - Not Yet Assigned	Glomerulonephritis (GN)	GN - Not Yet Assigned
CKD - Not Yet Assigned	2.05e-19	N/A	N/A	N/A
Glomerulonephritis (GN)	1.63e-177	4.78e-58	N/A	N/A
GN - Not Yet Assigned	2.07e-77	2.18e-28	1.54e-01	N/A
Pre-CKD	2.71e-08	3.12e-01	2.13e-44	1.27e-25